What do long-term bird counts tell us about bird population trends in the Triangle of NC?

Background and methods in brief

- a few sentences about background to explain why we do this analysis
- a short paragraph about what was done to compile and analyze the data (e.g. R scripts)

The New Hope Birds Project aims to study bird populations within the North Carolina Research Triangle. Our goal is/was to analyze trends in bird populations over several decades in various locations within the Research Triangle. In this, we can identify patterns pertaining to North Carolina birds based on their species and habitat types. In this study, we looked at Durham, Chapel Hill, and Jordan Lake as locations of interest.

To format and analyze the data, we used R as our primary language. First, we had to aggregate the Durham spring and winter excel sheets into two csv's for each season to match the Jordan Lake and Chapel Hill data formatting. We did this by excluding any columns that didn't give us any counts of bird populations. We then standardized each yearly data set and checked for duplicate species among the datasets. We then used a full join to combine all the datasets by species name, leaving us with only the counts for each species. To develop the habitat data spreadsheets, we grouped the species by habitats (Forest, grassland, aridland, wetland and coast, tundra, and habitat generalist). We mapped each bird species to habitats based on this species guide using a JSON file. From there, we calculated the population counts for each habitat type in an R file that pulled from the JSON formatting file.

Confounding variables

- Jordan Lake 2020 spring column is blank